

Thoughts on the analysis of the violin

This document has an in-depth explanation of all the math and file formats I use in the ObieApp.

1 FRFs or TRFs

We want to express the violin mathematically, we can view it as a transfer function (also called the frequency response function)- it converts some input (in this case a force from the hammer) to some output (in this case sound or a pressure fluctuation):

$$f(t) \rightarrow p(t) \quad (1)$$

or taking the Fourier Transform, we can find the transfer function for the violin as:

$$H(f) = \frac{P(f)}{F(f)} \quad (2)$$

where $P(f)$ and $F(f)$ are the Fourier transforms of pressure and force and have real and imaginary parts. Remember that the Fourier Transform is defined as:

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-i2\pi ft} dt \quad (3)$$

I can simplify - get the denominator to be real, by multiplying by F^* :

$$H(f) = \frac{P(f)}{F(f)} = \frac{PF^*}{FF^*} = H1 = \frac{PP^*}{FP^*} = H2 \quad (4)$$

2 Least Squares Fit

In reality, we take 5-10 hits per position so that we can get a “best fit” reading. We know that:

$$P(f) = H(f)F(f) \quad (5)$$

So the error of the best fit would be looking at the measured values (F_i and P_i):

$$\epsilon_i(f) = P_i - H(f)F_i \quad (6)$$

Now realizing that $H(f)$ is a complex number ($a + ib$) and that F, P are complex as well, gives (remember that the magnitude of the product equals the product of the magnitudes):

$$\begin{aligned} \epsilon_i^2(f) &= (P_i - H(f)F_i)(P_i - H(f)F_i)^* \\ &= |P_i|^2 - H(f)F_iP_i^* - P_iH^*(f)F_i^* + |H(f)F_i|^2 \\ &= |P_i|^2 - (a + ib)F_iP_i^* - P_i(a - ib)F_i^* + |(a + ib)F_i|^2 \\ &= |P_i|^2 - (a + ib)F_iP_i^* - P_i(a - ib)F_i^* + (a^2 + b^2)F_iF_i^* \end{aligned} \quad (7)$$

Next we sum over all the readings and then find the minimum by differentiating with respect to a and b and setting them equal to 0 to get:

$$\begin{aligned}\frac{\partial \sum \epsilon_i^2(f)}{\partial a} &= 0 = 0 - \sum F_i P_i^* - \sum P_i F_i^* + 2a \sum F_i F_i^* \\ \frac{\partial \sum \epsilon_i^2(f)}{\partial b} &= 0 = 0 - i \sum F_i P_i^* + i \sum P_i F_i^* + 2b \sum F_i F_i^*\end{aligned}\tag{8}$$

multiplying the second one by i and then adding you get:

$$\begin{aligned}0 &= - \sum F_i P_i^* - \sum P_i F_i^* + 2a \sum F_i F_i^* \\ 0 &= \sum F_i P_i^* - \sum P_i F_i^* + 2bi \sum F_i F_i^*\end{aligned}\tag{9}$$

I can add them together and get:

$$\begin{aligned}0 &= -2 \sum P_i F_i^* + 2a \sum F_i F_i^* + 2bi \sum F_i F_i^* \\ \sum P_i F_i^* &= H(f) \sum F_i F_i^*\end{aligned}\tag{10}$$

or

$$H_1(f) = \frac{\sum P_i F_i^*}{\sum F_i F_i^*}\tag{11}$$

Likewise we could look for reducing the effect of noise on the input – or using, as the error,

$$\epsilon_i(f) = F_i - \frac{P_i}{H(f)}\tag{12}$$

And do the analysis again and you will get:

$$\begin{aligned}\epsilon_i^2(f) &= (F_i - \frac{P_i}{H(f)})(F_i - \frac{P_i}{H(f)})^* \\ &= F_i F_i^* - F_i \frac{P_i^*}{H^*} - F_i^* \frac{P_i}{H} + \frac{P_i^* P_i}{H^* H} \\ &= F_i F_i^* - F_i P_i^* (a - ib)^{-1} - F_i^* P_i (a + ib)^{-1} + P_i P_i^* (a^2 + b^2)^{-1}\end{aligned}\tag{13}$$

Again summing and differentiating:

$$\begin{aligned}\frac{\partial \sum \epsilon_i^2(f)}{\partial a} &= 0 = 0 + (a - ib)^{-2} \sum F_i P_i^* + (a + ib)^{-2} \sum P_i F_i^* - (a^2 + b^2)^{-2} 2a \sum P_i P_i^* \\ \frac{\partial \sum \epsilon_i^2(f)}{\partial b} &= 0 = 0 - i(a - ib)^{-2} \sum F_i P_i^* + i(a + ib)^{-2} \sum P_i F_i^* - (a^2 + b^2)^{-2} 2b \sum P_i P_i^*\end{aligned}\tag{14}$$

multiplying the second equation through by i gives you:

$$\begin{aligned} 0 &= (a - ib)^{-2} \sum F_i P_i^* + (a + ib)^{-2} \sum P_i F_i^* - (a^2 + b^2)^{-2} 2a \sum P_i P_i^* \\ 0 &= (a - ib)^{-2} \sum F_i P_i^* - (a + ib)^{-2} \sum P_i F_i^* - (a^2 + b^2)^{-2} 2ib \sum P_i P_i^* \end{aligned} \quad (15)$$

Adding them together and reshuffling:

$$\begin{aligned} H(a^2 + b^2)^{-2} \sum P_i P_i^* &= (a - ib)^{-2} \sum F_i P_i^* \\ H \sum P_i P_i^* &= \frac{(a + ib)^2 (a - ib)^2}{(a - ib)(a - ib)} \sum F_i P_i^* \\ \sum P_i P_i^* &= H \sum F_i P_i^* \\ H_2(f) &= \frac{\sum P_i P_i^*}{\sum F_i P_i^*} \end{aligned} \quad (16)$$

gives us the best fit in the other direction (assuming that the input is noisy). If the data is clean and the noise is truly random, these two values for H should be very close and therefore their ratio, the coherence, should be 1.

$$C = \frac{H_1}{H_2} \quad (17)$$

If it is not 1, then one (or both) of your signals is noisy (not a great least squares fit). George then used the geometric mean.

$$H \sim \sqrt{H_1 H_2} \quad (18)$$

3 Claude's explanation

Coherence is essentially a measure of how consistently a system behaves across repeated measurements.

Think of it this way: you tap a violin with a hammer and measure the input force and the output sound. If you tap it again with a slightly different force, does the ratio of output to input stay the same? If the system is well-behaved (linear, no noise), it should — and the coherence will be close to 1. If something is messy (nonlinearity, background noise, a loose part rattling), the ratio will vary between hits, and coherence drops toward 0. More concretely, coherence answers: "how much of the output can be explained by the input?" The formula from the document makes this intuitive:

$$C = \frac{|\sum P_i F_i^*|^2}{\sum F_i F_i^* \sum P_i P_i^*} \quad (19)$$

The numerator is the cross-spectrum squared — how much F and P move together. The denominator is the product of the individual power spectra — how much each signal varies on its own. It’s essentially a frequency-domain analog of R^2 from linear regression. A few practical rules of thumb:

- $C \sim 1$: the output is well explained by the input — clean measurement, linear system
- $C \sim 0$: the output has little to do with the input at that frequency — probably noise, or the system simply doesn’t respond there
- C between 0 and 1: partial explanation — could be noise, nonlinearity, or multiple inputs contributing to the output

One important subtlety: coherence is frequency-dependent. A violin might have $C \sim 1$ near a resonance (where it responds strongly and predictably) but low coherence in a frequency range where it barely responds and the output is dominated by noise. So you’d typically plot it across the whole frequency range rather than summarize it as a single number.

4 File Formats

4.1 TRF files

Parse and build binary `.trf` files (Rational Acoustics / MtxVec format).

Header Format

Little-endian, 110 bytes total:

Type	Fields
I	index
4d	Xr, Yr, Xactual, YActual
2s	char_str
3d	Hz_Resolution, Start_Freq, End_Freq
11f	fComplex, fLength, ... (9 unused floats)
4s	caption

Data (from byte 110)

- **Complex:** alternating float64 real/imag pairs
- **Real:** plain float64 values

4.2 File Formats

4.2.1 AvR / AvC File Format Specification

AvR files store a *real-valued* averaged spectrum; AvC files store a *complex-valued* averaged spectrum. The file layout is identical except for the data section, which is $N \times 8$ bytes for AvR and $N \times 16$ bytes (interleaved Re/Im pairs) for AvC, where N is given by **fLength** in the TSingleArray descriptor. Type conventions: fields prefixed **f** are Single-precision floats (4 bytes); all other 4-byte TSingleArray fields are U32.

Field	Bytes	Type	Description / Possible Values
<i>Header</i>			
DataType	1	TDataType	Response quantity (see enum below).
HzRes	8	Double	Frequency resolution Δf (Hz / bin). > 0 .
StartFrequency	8	Double	Start frequency (Hz). ≥ 0 .
EndFrequency	8	Double	End frequency (Hz). $> \text{StartFrequency}$.
ScaleFactor	8	Double	Scale factor applied to the averaged data.
nAverages	4	U32	Number of averages used. Typically 1–1000.
AverageType	1	Enum	Averaging method (see enum below).
<i>AverageType</i> enum values			
RMS	—	value 0	Root-mean-square averaging.
Mean	—	value 1	Linear (arithmetic mean) averaging.
Complex	—	value 2	Complex (vector) averaging.
Geometric	—	value 3	Geometric mean averaging.
None	—	value 4	No averaging (single measurement).
<i>TSingleArray</i> (MtxVec array descriptor — all fields 4 bytes)			
fComplex	4	Single	0.0 = real (AvR), 1.0 = complex (AvC).
fLength	4	Single	Number of data points N (stored as float). Determines data section size.

Field	Bytes	Type	Description / Possible Values
a[2]	4	U32	Reserved / alignment padding.
fConditionCheck	4	Single	Internal integrity / validation value.
Precision	4	U32	0 = Single (32-bit), 1 = Double (64-bit).
MtxVecVersion	4	U32	Version of MtxVec library that wrote the file.
SizeOf(TSample)	4	U32	Byte size of each data sample.
Tag	4	U32	User-defined tag or identifier.
MtxVecFileCode	4	U32	Magic number identifying this as a MtxVec file.
a[9]	4	U32	Reserved / alignment padding.
a[10]	4	U32	Reserved / alignment padding.
caption	4	String	Short label, 4 bytes, null-padded (\00\00\00\00 if empty).
<i>Data</i> (length determined by fLength and fComplex)			
AvR data	$N \times 8$	Double[]	N real-valued doubles, little-endian. Present when fComplex = 0.
AvC data	$N \times 16$	Double[]	N complex samples as interleaved Re/Im double pairs ($2N$ doubles total), little-endian. Present when fComplex = 1.
<i>Notes / Comments</i> (appended after data section)			
notes	var	String	Free-text annotation. Starts at byte offset $N \times 8$ (AvR) or $N \times 16$ (AvC) from the beginning of the data section, i.e. after fLength or $2 \times \text{fLength}$ doubles respectively.

Data section size summary:

$$\text{AvR: } N \times 8 \text{ bytes} \quad \text{AvC: } N \times 16 \text{ bytes} \quad \text{where } N = \text{int}(\text{fLength})$$

Notes: (1) f-prefixed TSingleArray fields are Single-precision floats (4 bytes). (2) MtxVecFileCode and SizeOf(TSample) are U32. (3) caption is 4 bytes, null-padded. (4) All multi-byte values are little-endian. (5) TDataType is stored as a single byte (U8). Source: george.pas (Dew Research / DSP Master) and LabVIEW writer.

4.2.2 FRF File Format Specifications

Fields verified against the Delphi source (`george.pas`) and LabVIEW writer code. Type conventions: fields prefixed `f` are Single-precision floats (4 bytes); `MtxVecFileCode` and `SizeOf(TSample)` are U32; `caption` is a null-padded string.

Field	Bytes	Type	Description / Possible Values
<i>TFRFHeader</i> (Delphi packed record — no padding between fields)			
index	4	Integer	Channel or measurement index. ≥ 0 .
Xr	8	Double	Reference (input) channel number or physical X location.
Yr	8	Double	Response (output) channel number or physical Y location.
XActual	8	Double	Physical X-axis value (e.g. excitation location).
YActual	8	Double	Physical Y-axis value (e.g. response location).
FRFType	1	TFRFType	0 = <code>ftTransfer (.trf)</code> , 1 = <code>ftPoint (.pnt)</code> .
DataType	1	TDataType	Response quantity (see enum below).
HzRes	8	Double	Frequency resolution Δf (Hz / bin). > 0 .
StartFrequency	8	Double	Start frequency (Hz). ≥ 0 .
EndFrequency	8	Double	End frequency (Hz). $> \text{StartFrequency}$.
<i>TDataType</i> enum values (U8)			
dtAccel	—	value 0	Accelerance (acceleration / force).
dtMob	—	value 1	Mobility (velocity / force).
dtRecept	—	value 2	Receptance (displacement / force).
dtMic	—	value 3	Microphone response.
<i>TSingleArray</i> (MtxVec internal array descriptor — all fields 4 bytes)			
fComplex	4	Single	Array-level complex flag: 0.0 = real, 1.0 = complex.
fLength	4	Single	Number of elements in the data array (stored as float).
a[2]	4	U32	Reserved / alignment padding.
fConditionCheck	4	Single	Internal integrity / validation value.
Precision	4	U32	0 = Single (32-bit), 1 = Double (64-bit).
MtxVecVersion	4	U32	Version of MtxVec library that wrote the file.
SizeOf(TSample)	4	U32	Byte size of each data sample.
Tag	4	U32	User-defined tag or identifier.

Field	Bytes	Type	Description / Possible Values
MtxVecFileCode	4	U32	Magic number identifying this as a MtxVec file.
a[9]	4	U32	Reserved / alignment padding.
a[10]	4	U32	Reserved / alignment padding.
<i>Payload</i>			
caption	4	String	Short label, 4 bytes, null-padded (\00\00\00\00 if empty).
FRF data	$N \times 16$	Double[]	Complex FRF samples as interleaved Re / Im DBL pairs, little-endian. N = number of frequency lines.

5 Adiabatic Wave Equation

If the fluid is adiabatic, then

$$p = K\rho^\gamma$$

so

$$\frac{\partial p}{\partial x} = K\gamma\rho^{\gamma-1}\frac{\partial \rho}{\partial x} = c^2\frac{\partial \rho}{\partial x}$$

and

$$\frac{\partial p}{\partial t} = K\gamma\rho^{\gamma-1}\frac{\partial \rho}{\partial t} = c^2\frac{\partial \rho}{\partial t}$$

But from continuity

$$\frac{\partial \rho}{\partial t} + \rho_0\frac{\partial u}{\partial x} = 0$$

so

$$\frac{\partial p}{\partial t} + \frac{\rho_0}{c^2}\frac{\partial u}{\partial x} = 0$$

$$\frac{\partial p}{\partial t} = -\frac{\rho_0}{c^2}\frac{\partial^2 y}{\partial x \partial t}$$

or integrating by time,

$$p = -\frac{\rho_0}{c^2}\frac{\partial y}{\partial x} = -\kappa\frac{\partial y}{\partial x}$$

where κ is the adiabatic bulk modulus

and

$$\rho_0\frac{\partial u}{\partial t} + \frac{\partial p}{\partial x} = 0$$

or

$$\frac{\partial p}{\partial x} = -\rho_0\frac{\partial^2 y}{\partial t^2}$$

and

$$p = K\rho^\gamma$$

so

$$\frac{\partial p}{\partial x} = K\gamma\rho^{\gamma-1}\frac{\partial \rho}{\partial x} = c^2\frac{\partial \rho}{\partial x}$$

5.1 wave equation

Assume that the pressure is p and the speed of a fluid particle is u with a mean of zero and the density everywhere is $\rho_0 + \rho$. Then the 1D continuity gives us:

$$\frac{\partial(\rho_0 + \rho)}{\partial t} + \frac{\partial(\rho_0 + \rho)u}{\partial x} = 0$$

or

$$\frac{\partial \rho}{\partial t} + \rho_0 \frac{\partial u}{\partial x} + \frac{\partial(\rho u)}{\partial x} = 0$$

and assuming the last term is small compared to the second to last, you get:

$$\frac{\partial \rho}{\partial t} + \rho_0 \frac{\partial u}{\partial x} = 0$$

or

$$\frac{\partial^2 \rho}{\partial t^2} + \rho_0 \frac{\partial^2 u}{\partial x \partial t} = 0$$

The momentum equation (1D, no gravity or friction terms) is:

$$\frac{\partial((\rho_0 + \rho)u)}{\partial t} + (\rho_0 + \rho)u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0$$

or

$$\rho_0 \frac{\partial u}{\partial t} + \rho \frac{\partial u}{\partial t} + u \frac{\partial \rho}{\partial t} + \rho_0 u \frac{\partial u}{\partial x} + \rho u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0$$

Assuming the fluctuations are small:

$$\rho_0 \frac{\partial u}{\partial t} + \rho_0 u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0$$

Differentiate with respect to x :

$$\rho_0 \frac{\partial^2 u}{\partial t \partial x} + \rho_0 u \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 p}{\partial x^2} = 0$$

Dropping the middle term since derivatives in u should be about the same in both x and t and therefore the first term should be a lot bigger than the second - and combining both equations you get:

$$\frac{\partial^2 p}{\partial x^2} = \frac{\partial^2 \rho}{\partial t^2}$$

If we assume an adiabatic process, then

$$p = K \rho^\gamma$$

So

$$\frac{\partial p}{\partial t} = \gamma K \rho^{\gamma-1} \frac{\partial \rho}{\partial t}$$

or

$$\frac{\partial^2 p}{\partial t^2} = \gamma K \rho^{\gamma-1} \frac{\partial^2 \rho}{\partial t^2} + \gamma(\gamma-1) K \rho^{\gamma-2} \left(\frac{\partial \rho}{\partial t}\right)^2$$

Linearizing loses the last term:

$$\frac{\partial^2 p}{\partial t^2} = \gamma K \rho^{\gamma-1} \frac{\partial^2 \rho}{\partial t^2}$$

or

$$\frac{\partial^2 p}{\partial t^2} = \gamma \frac{p}{\rho^\gamma} \rho^{\gamma-1} \frac{\partial^2 \rho}{\partial t^2}$$

or

$$\frac{\partial^2 p}{\partial t^2} = \gamma RT \frac{\partial^2 \rho}{\partial t^2}$$

Combining then gives

$$\frac{\partial^2 p}{\partial x^2} = \frac{\partial^2 p}{\partial t^2} \frac{1}{\gamma RT}$$

or

$$\frac{\partial^2 p}{\partial t^2} = \gamma RT \frac{\partial^2 p}{\partial x^2}$$

5.2 Acoustic Impedance

Next, we want to look at the acoustic impedance of the plate (acoustic impedance, Z , is the ratio of pressure to volume flow, specific acoustic impedance, z , is the ratio of pressure to volume flow per unit area - or acoustic flow velocity).

$$p = ZU$$

For an infinitely long pipe, cross-sectional area S , we get:

$$Z = \frac{p}{U} = \frac{F}{SU} = \dots = \frac{\rho c}{S}$$

- I do not see it

Defining specific acoustic impedance to be:

$$p = zu$$

where a change in pressure (p) drives the fluid (u) like a voltage drives a current.
then

$$\frac{F/A}{u} = z$$

We can look at the forces on the plate, there should be the force due to pressure, a spring-like restoring force and a damping force due to the fact that the material is hardly a perfect spring:

$$\sum F = ma = pA - kx - \beta u$$

or

$$z = \frac{ma + kx + \beta u}{Au} = \frac{m}{A} \frac{\ddot{x}}{\dot{x}} + \frac{\beta}{A} + \frac{k}{A} \frac{x}{\dot{x}}$$

If we assume a solution of the form

$$u = \dot{x} = Ce^{i\omega t}$$

then the impedance becomes:

$$z = \frac{m}{A} \frac{i\omega C e^{i\omega}}{C e^{i\omega}} + \frac{\beta}{A} + \frac{k}{A} \frac{C e^{i\omega}/i\omega}{C e^{i\omega}}$$

$$z = \frac{m}{A} i\omega + \frac{\beta}{A} + \frac{k}{A} \frac{1}{i\omega}$$

or

$$z = \frac{\beta}{A} + i\omega \frac{m}{A} - \frac{i}{\omega \frac{A}{k}}$$

or

$$z = r + i\omega M - \frac{i}{\omega C}$$

where

$$\begin{aligned} r &= \frac{\beta}{A} \\ M &= \frac{m}{A} \\ C &= \frac{A}{k} \end{aligned}$$

5.3 Sound Intensity Measurements

Sound intensity is defined as the sound power per unit surface area. For a point source:

$$I = \frac{P}{4\pi r^2}$$

So the Sound Power Level is referenced to $P_0 = 10^{-12}W$ and the Sound Intensity Level is referenced to $I_0 = 10^{-12}W/m^2$

$$SWL = 10\log_{10}(P/P_0) \text{ and } SIL = 10\log_{10}(I/I_0)$$

Since

$$I = pu = \frac{p^2}{z}$$

then the sound pressure level is:

$$SPL = 10\log_{10}(p^2/p_0^2) = 20\log_{10}(p/p_0)$$

Sound hole is modeled with a frictional resistance R_h :

$$m_{hole-air} \ddot{x}_h = A_h \Delta p - R_h \dot{x}_h$$

and the top plate has a restoring force:

$$m_{plate} \ddot{x} = F - k_p x_p - R_p \dot{x}_p + A_p \Delta p$$

5.4 Quality factor

Usually defined as

$$Q = \frac{f_{center}}{\Delta f} = \frac{\sqrt{mk}}{R} = \frac{\omega_0}{2r}$$

where the spread of the peak (Δf) is typically measured by the 1/2 power reduction point, r is the exponential decay of the signal and R is the damping coefficient ($m\ddot{x} + R\dot{x} + kx = 0$)